

REV2021

SEMESTER 2

3 CREDITS

Textile Fibres

Course Code: 2061 | Program Core

[Official Syllabus Source](#)

Course Overview

Teaching Scheme

- Lectures: 2 Hours/Week
- Tutorials: 1 Hour/Week
- Total Periods: 45 Hours

Assessment Scheme

- Continuous Internal Evaluation (CIE)
- End Semester Examination (ESE)
- Two Series Tests

This course provides foundational knowledge on natural vegetable, animal, and semi-synthetic fibres. It covers their chemical structure, physical properties, and industrial applications, with a specific focus on cotton, silk, wool, and synthetic polyamides like Nylon 6 and 6,6.

Course Outcomes (COs)

CO No.	Description	Cognitive Level
CO1	Identify classification and general properties of textile fibres. Illustrate structure, properties & uses of Cotton.	Understanding
CO2	Describe structure, properties and uses of Bast fibres (Linen, Jute) & Animal fibres (Silk, Wool).	Understanding
CO3	Explain manufacturing process, structure, properties and uses of Semi-synthetic fibres.	Understanding
CO4	Explain types of polymerizations and manufacturing process, properties and uses of Nylon 6.6 & Nylon 6.	Understanding

Module 1 | 11 Hours

Classification & Cotton Fibre

1.1 Classification of Textile Fibres

Textile fibres are the fundamental units of textile structures. They are classified based on their **Origin** and **Chemical Nature**.

Classification by Origin

- **Natural Fibres:**
 - Vegetable (Cellulosic): Cotton, Linen, Jute, Hemp.
 - Animal (Protein): Silk, Wool.
 - Mineral: Asbestos.
- **Man-made Fibres:**
 - Regenerated (Semi-synthetic): Viscose Rayon, Acetate Rayon.

- Synthetic: Nylon, Polyester, Acrylic.
- Inorganic: Glass, Metallic fibres.

Malayalam Note: ഫൈബറുകൾക്ക് രണ്ടു തരം ഉത്ഭവമുണ്ട്. സ്വാഭാവികമായി ഉണ്ടാകുന്നവയെ (Natural) ഫൈബറുകൾ എന്ന് വിളിക്കുന്നു. മനുഷ്യൻ ഉണ്ടാക്കിയവയെ (Man-made) ഫൈബറുകൾ എന്ന് വിളിക്കുന്നു. കോട്ടൺ (Cotton), സിൽക്ക് (Silk), വൂൾ (Wool) സ്വാഭാവിക ഫൈബറുകളാണ്. നയ്ലോൺ, പോളിസ്റ്റർ, അക്രിലിക്, ഗ്ലാസ്, ലോഹ ഫൈബറുകൾ മനുഷ്യൻ ഉണ്ടാക്കിയവയാണ്.

1.2 Cotton: Structure & Properties

Cotton is a natural vegetable fibre obtained from the seed hair of the *Gossypium* plant.

Structure of Cotton

Longitudinal View (Convolutions)



Cross Section (Kidney

Fig: Longitudinal view showing natural twists (convolutions) and kidney-shaped cross-section.

Physical Properties

- **Length:** 1.25 cm to 6 cm (Staple fibre).
- **Strength:** Increases when wet (unique property).
- **Fineness:** Varies by variety (Sea Island is finest).
- **Colour:** Usually white to creamy.

Chemical Properties

- **Effect of Acid:** Disintegrated by hot dilute or cold concentrated mineral acids.
- **Effect of Alkali:** Resistant to alkalis; swelling occurs in strong NaOH (Mercerization).
- **Microorganisms:** Attacked by mildew and fungi in damp conditions.

Module 2 | 11 Hours

Bast & Animal Fibres

2.1 Bast Fibres: Linen and Jute

Bast fibres are obtained from the stem (phloem) of plants.

Linen (Flax)

Extraction: Processed via *Retting* (bacterial action in water to loosen fibres from the stem) followed by Scutching and Hackling.

Properties: Stronger than cotton, high luster, good heat conductor (cool to wear).

Jute

Known as the "Golden Fibre". It is cheap, strong, but coarse and harsh. Used mainly for packaging (gunny bags) and carpet backing.

2.2 Animal Fibres: Silk and Wool

Silk (The Queen of Fibres)

Sericulture: The commercial rearing of silkworms (*Bombyx mori*) for the production of silk.

Lifecycle: Egg → Larva (Caterpillar) → Pupa (in Cocoon) → Moth.

Degumming: Removing the silk gum (Sericin) using soap and water to reveal the lustrous fibroin filaments.

Wool

Scouring: The process of washing raw wool to remove grease (lanolin), dirt, and dried sweat (suint).

Carbonisation: Treatment with acid to remove vegetable matter (burrs, seeds) from wool.

Felting: The interlocking of wool scales under heat, moisture, and pressure.

Module 3 | 10 Hours

Semi-Synthetic Fibres

3.1 Viscose Rayon

Viscose Rayon is regenerated cellulose. It is made from wood pulp or cotton linters.

Manufacturing Flowchart

Wood Pulp → Steeping (NaOH) → Shredding (Alkali Cellulose) → Ageing → Xanthation (CS₂) → Dissolving → Ripening → Spinning (Wet Spinning) → Regeneration.

Properties

- Highly absorbent.
- Loses 50% strength when wet.
- Good drape and soft feel.

3.2 Modern Semi-Synthetics



Module 4 | 11 Hours

Synthetic Fibres & Polymerization

4.1 Principles of Polymerization



Monomer: The small repeating unit.

Polymer: A long chain molecule made of many monomers.

Types of Polymerization

1. **Addition Polymerization:** Monomers join without loss of any small molecules (e.g., Polyethylene).
2. **Condensation Polymerization:** Monomers join with the elimination of small molecules like water or ammonia (e.g., Nylon 6,6).

4.2 Polyamides: Nylon 6,6 and Nylon 6

Nylon 6,6

Made from **Hexamethylene diamine** and **Adipic acid**. Both monomers have 6 carbon atoms, hence the name "6,6".

Nylon 6

Made from a single monomer: **Caprolactam**. It involves ring-opening polymerization.

Spinning Methods

Method	Mechanism	Example
Melt Spinning	Polymer is melted and extruded into cold air.	Nylon, Polyester
Dry Spinning	Polymer dissolved in solvent; solvent evaporated by hot air.	Acetate, Acrylic
Wet Spinning	Polymer solution extruded into a chemical coagulating bath.	Viscose Rayon

Formula Bank & Calculations

Draw Ratio Calculation

Drawing is the process of stretching the extruded filament to align polymer chains and increase strength.

Formula: $\text{Draw Ratio} = L_{\text{final}} / L_{\text{initial}}$

Example: A nylon filament of length 10cm is stretched to 45cm. Calculate the draw ratio.

Given: $L_{\text{initial}} = 10\text{cm}$, $L_{\text{final}} = 45\text{cm}$

Calculation: $45 / 10 = 4.5$

Answer: Draw Ratio is 4.5:1.

Interactive Draw Ratio Calculator

Initial Length (m):

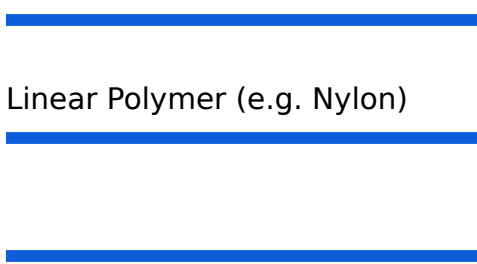
e.g. 1

Final Length (m):

e.g. 4

Calculate Ratio

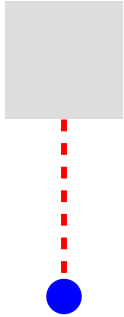
Diagram Library



Linear Polymer (e.g. Nylon)

The diagram shows three parallel horizontal blue lines, representing the structure of a linear polymer.

Linear Polymer Structure



Melting

The diagram shows a grey square at the top, a red dashed line extending downwards, and a blue circle at the bottom. The red dashed line is labeled 'Extrusion'.

Extrusion

Melt Spinning Schematic

Module-wise Practice Questions

► 1. Define Textile Fibre. (2 Marks)

► 2. Differentiate between Nylon 6 and Nylon 6,6. (5 Marks)

► 3. Explain the process of 'Retting' in Linen production. (3 Marks)

Practice Model Paper

Time: 3 Hours | Max Marks: 75

Part A (Answer all questions - 10 x 2 = 20 Marks)

1. What is a staple fibre?
2. Define Sericulture.
3. What is the chemical composition of cotton?
4. Name two bast fibres.

5. What is the monomer of Nylon 6?
6. Define Polymerization.
7. What is Scouring?
8. List two uses of Jute.
9. What is the function of a spinneret?
10. Name a solvent used in dry spinning.

Part B (Answer any 5 - 5 x 5 = 25 Marks)

1. Explain the classification of fibres with a chart.
2. Describe the physical properties of Silk.
3. Explain the manufacturing of Viscose Rayon.
4. Compare Melt spinning and Wet spinning.
5. Explain the effect of acids and alkalis on Cotton.

Part C (Answer any 3 - 3 x 10 = 30 Marks)

1. Explain the structure, properties, and uses of Wool in detail.
2. Describe the manufacturing process of Nylon 6,6 with a flowchart.
3. Discuss the extraction and properties of Linen.

Quick Revision

Fibre	Type	Key Feature
Cotton	Natural Vegetable	Convolution, Kidney cross-section.
Silk	Natural Animal	Continuous filament, Sericin gum.
Wool	Natural Animal	Surface scales, Felting property.
Viscose	Regenerated Cellulose	High absorbency, low wet strength.

Nylon	Synthetic Polyamide	High strength, thermoplastic.
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Glossary

Convolution:

The natural twist found in cotton fibres.

Lumen:

The central hollow canal in a cotton fibre.

Thermoplastic:

A material that softens when heated and hardens when cooled.

Spinneret:

A metal plate with tiny holes through which polymer dope is extruded to form filaments.

Official Source Declaration

This content is based on the **SITTTR Kerala Revision 2021** syllabus for Course 2061 (Textile Fibres). The structure, module distribution, and learning outcomes are derived directly from the official document.

References:

- V.A. Shenai, Textile Fibres - Vol. 1, 2
- S.P. Mishra, Fibre Science and Technology
- J. Gordon Cook, Handbook of Textile Fibres

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- **Module 2:** □□□□, □□□ (Jute) □□□□□□□□ □□□□□□□□□□□□ □□□□□, □□□□□□□ □□□□□□□□□□□□.
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